

INTEREST

SIMPLE INTEREST

It is calculated on the basis of a basic amount borrowed for the entire period at a particular rate of interest.

The amount borrowed is the principal for the entire period of borrowing.

$$SI = \frac{P \times R \times T}{100}$$

where, **P** is Principal

R is % rate of interest

T is the time duration.

Amount **A** is the sum of Principal **P** and **SI** so

$$A = P + SI$$

COMPOUND INTEREST

The interests of the previous years are added to the principal for the calculation of the compound interest.

$$CI = P [1 + R/100]^T - P$$

where, P is the Principal Amount,

R is the Rate of Interest Per Annum,

T is the Time duration in Years.

IMPORTANT FORMULA

$$\text{Successive Rate of Interest} = A + B + \frac{AB}{100}$$


COMPARISON BETWEEN SIMPLE INTEREST AND COMPOUND INTEREST

Let, Principal = Rs. 100 and Rate of Interest Per Year = 10 %

	SIMPLE INTEREST $SI = \frac{P \times R \times T}{100}$	COMPOUND INTEREST $CI = P [1 + R/100]^T - P$
1 st Year	10	10
2 nd Year	10	11
3 rd Year	10	12.1
4 th Year	10	13.31

Thus, we can conclude that every year Simple Interest remains constant as it is calculated on the Principal value where as the Compound Interest increases as it is calculated on the Amount.

Case 1: When interest is **compounded annually** then total amount A will be

$$A = P[1 + R/100]^T$$

Case 2: When interest is compounded half yearly then $A = P \left[1 + \frac{\frac{R}{2}}{100} \right]^{2T}$

Case 3: When interest is compounded quarterly then $A = P \left[1 + \frac{\frac{R}{4}}{100} \right]^{4T}$

Case 4: When differential rate of interest is charged i.e. if rate of interest is

R_1 % for the first year, R_2 % for the second year, R_3 % for the third year then

$$\text{Amount } A = P \left[1 + \frac{R_1}{100} \right] \left[1 + \frac{R_2}{100} \right] \left[1 + \frac{R_3}{100} \right]$$

1. The simple interest on a certain sum of money at 5% per annum for 4 years is ₹200 more than the interest on the sum for 3 years at 6% per annum. Find the sum.

A. 20000

B. 10000

C. 15000

D. 18000



2. In how many years does a sum of money become 4 times at the simple interest rate of 10% p.a?

A. 30 years

B. 45 years

C. 60 years

D. 90 years

3. The rate of interest for the first 3 years is 5% p.a for the next 3 years is 8% and beyond this, it is 10%. If the simple interest for 10 years is ₹4977 then find out the principal.

A. 8485

B. 7435

C. 6845

D. 6300



4. A sum of money put out on simple interest doubles in itself 22.5 years. In how many years would it triple itself?

A. 25 years

B. 45 years

C. 65 years

D. None



5. Simple interest on a sum of money is $\frac{1}{16}$ th of the principal, and rate percent per annum is equal to the number of years. Then the rate per cent is:

A. 2.5%

B. 5%

C. 5.5%

D. 7%

FV=



6. A sum was put at simple interest at a certain rate for 4 years. Had it been put at 2% higher rate, it would have fetched ₹200 more. Find the sum.

A. 5800

B. 4500

C. 2500

D. 1200



7. The annual payment that will discharge a debt of ₹1200 in 3 years at 10% per annum is.

A. 252.50

B. 363.63

C. 433.85

D. 486.74

The formula for the future value of an ordinary annuity is

$$FV = P + P \left(1 + \frac{R}{100}\right) + P \left(1 + \frac{R}{100}\right)^2 + \dots \dots P \left(1 + \frac{R}{100}\right)^{T-1}$$

8. After what time will the sum of ₹10000 become ₹12000 at 5% per annum simple interest?

A. 3 years

B. 2 years

C. 5 years

D. 4 years



9. A sum of ₹9000 amounts to ₹10500 in 3 years at a certain rate of simple interest. In how many years at the same rate of simple interest the sum of will get doubled?

A. 6 years

B. 9 years

C. 15 years

D. 18 years



10. A sum of ₹9000 amounts to ₹12000 in 8 years at simple interest. If the rate per annum is further increased by 2%, the amount will be:

A. 10280

B. 12445

C. 13442

D. 14892



11. A sum of money placed at compound interest amounts to ₹6400 in 3 years and to ₹6800 in 4 years. Find the rate percent per annum.

A. $23/4$ %

B. $25/4$ %

C. $27/4$ %

D. None of these



12. Find the amount of ₹8000 for 1.5 years at 5% per annum, if the interest is compounded semi-annually.

A. 9632

B. 9142

C. 8615

D. 8321



13. If the compound interest on a sum of money for 2 years at 5% per annum is ₹100 more than the simple interest on the same sum for the same period and at the same rate. Find the sum.

A. 20000

B. 30000

C. 40000

D. 50000



14. By how much would the compound interest exceed the simple interest if the sum of ₹10000 were invested at 8% per annum for 2 years?

A. 56

B. 58

C. 67

D. 64



15. The difference between the compound interest and the simple interest for the sum ₹3000 at 10% p.a for 2 years is:

- A. 10 B. 20 C. 30 D. None of these**



16. Find the compound interest on ₹8000 for 4 years at 6% p.a?
- A. 3578.5 B. 2099.8 C. 1678.7 D. 1234.5

17. The population of a village increases by 10% every year. If present population of the village is 12000, what was the population of this village 4 years ago?

A. 9846

B. 8196

C. 7342

D. 6892



18. A certain sum of money has been placed at compound interest. If the compound interest for the two successive years be ₹12000 and ₹15600. Then find the rate of interest.

A. 10.2 %

B. 20.5 %

C. 30 %

D. 35 %



19. What will be the amount if a sum of ₹12400 is invested for 1 year at 10% per annum compound interest, interest being compounded half yearly?

A. ₹28931

B. ₹21461

C. ₹19421

D. ₹13671



20. At what rate of compound interest per annum will a sum of Rs. 1200 become Rs. 1348.32 in 2 years?

A. 6 %

B. 6.5 %

C. 7 %

D. 7.5 %

21. The principal amount that becomes Rs. 4913 in 3 years at interest rate of $25/4\%$ per annum when interest is compounded annually is:

A. 4306

B. 4396

C. 4096

D. 4809

22. The difference between the compound interest and simple interest on certain sum at 10% per annum for 2 years is Rs. 631. What is the sum?

A. 63100

B. 56000

C. 42160

D. 68000



23. A certain sum becomes 8 times itself in 3 years under compound interest, interest being compounded annually. How many years would the sum takes to become 64 times itself?

A. 6 years

B. 5 years

C. 4 years

D. 8 years



24. The least number of complete years in which a sum of money put out at 20% compound interest will be more than doubled is:

A. 3 years

B. 4 years

C. 5 years

D. 6 years



25. On a certain sum of money the compound interest Rs. 318 is earned in 2 years. If the rate of interest is 12%, what is the principal amount?

A. 1250

B. 1300

C. 1200

D. 1150



26. The simple interest on a certain sum of money at rate of interest 5% per annum for 2 years is Rs. 500. What is the compound interest on the same sum for the same period and at the same rate of interest?

A. 412.5

B. 400

C. 500

D. 512.5



27. In how many years the compound interest on Rs. 10000 at the rate of 10% will be Rs. 2100 ?

A. 1.5 years

B. 2 years

C. 2.5 years

D. 3 years



- 28. The difference between Simple Interest and Compound Interest on a certain sum of money for 2 years is Rs.200. If the rate of interest is 12.5%, what is the sum?**
- A. 11800 B. 12000 C. 12800 D. 13000**
- 

29. If the difference between Simple Interest and Compound Interest on a sum of money for 3 years is Rs. 200 and the rate of interest is 10%, what is the sum?

A. 6451.6

B. 6351.6

C. 6251.6

D. 6151.6



30. The difference between simple interest and compound interest on a sum of money for 1 year at 4% per 6 months is 4, what is the sum?

A. 2100

B. 2300

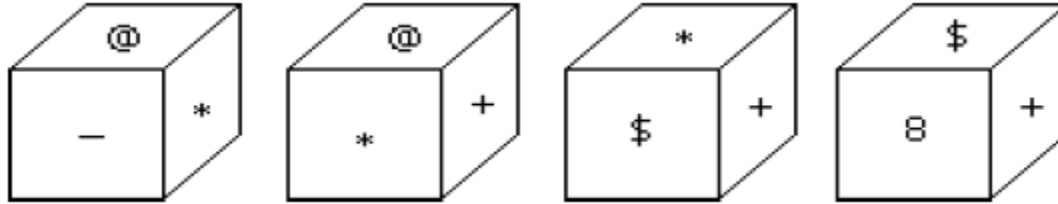
C. 2500

D. 3000



DICE

1. Which symbol will be on the face opposite to the face with symbol * ?
- A. @ B. \$ C. 8 D. +



Ans B \$

DICE

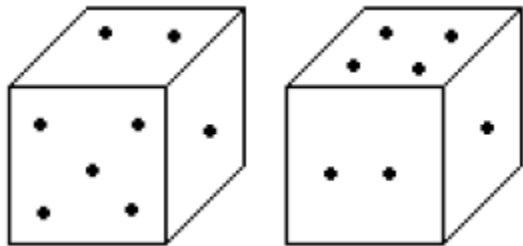
2. Two positions of dice are shown below. How many points will appear on the opposite to the face containing 5 points?

A. 3

B. 1

C. 2

D. 4



Ans D 4

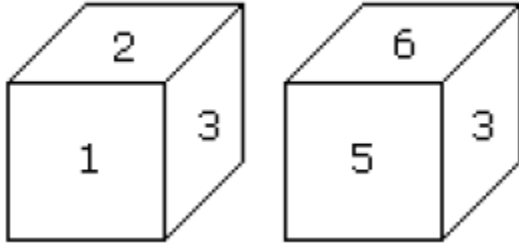
3. Which digit will appear on the face opposite to the face with number 4?

A. 3

B. 5

C. 6

D. 2/3



Ans A 3

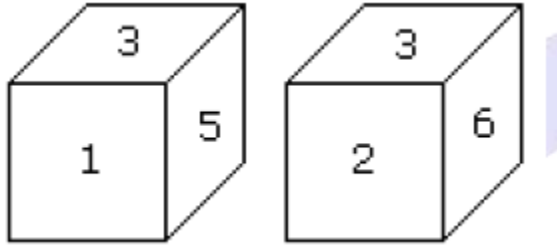
4. Two positions of a dice are shown below. Which number will appear on the face opposite to the face with the number 5?

A. 2/6

B. 2

C. 6

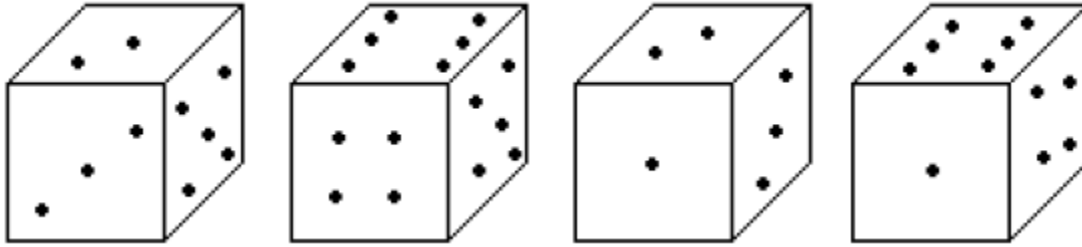
D. 4



Ans C 6

5. How many points will be on the face opposite to in face which contains 2 points?

A. 6



Ans D 6

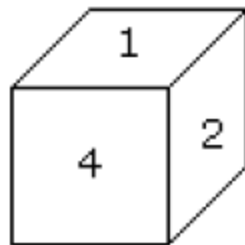
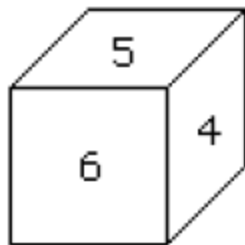
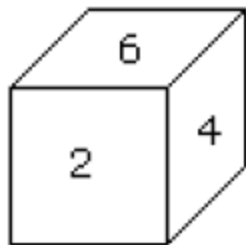
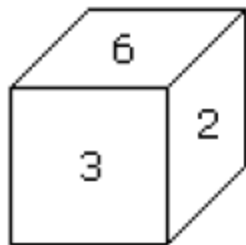
6. Which number is on the face opposite to 6?

A. 4

B. 1

C. 2

D. 3



Ans B 1

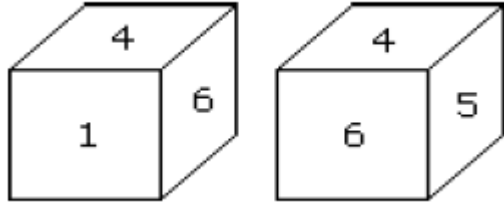
7. Two positions of a dice are shown below. When number '1' is on the top. What number will be at the bottom?

A. 3

B. 5

C. 2

D. 6



Ans B 5

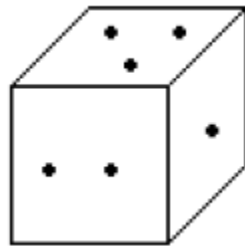
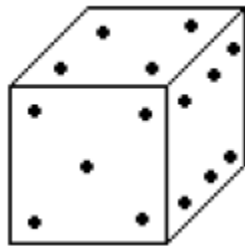
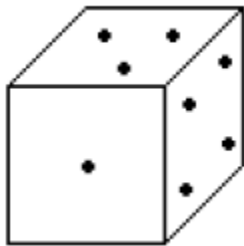
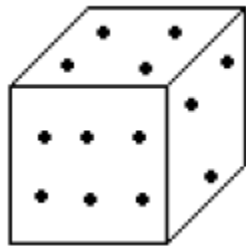
8. How many points will be on the face opposite to the face which contains 3 points?

A. 2

B. 4

C. 5

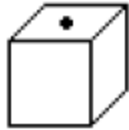
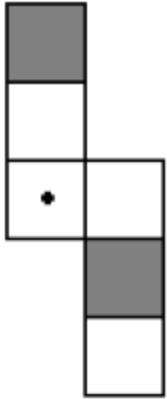
D. 6



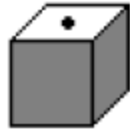
Ans C 5

9. The figure given on the left-hand side in each of the following questions is folded to form a box. Choose from the alternatives (1), (2), (3) and (4) the boxes that is similar to the box formed.

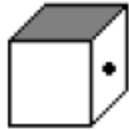
A. 2 & 3 only B. 1, 3 & 4 only C. 2 and 4 only D. 1 & 4 only



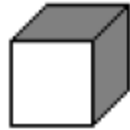
(1)



(2)



(3)

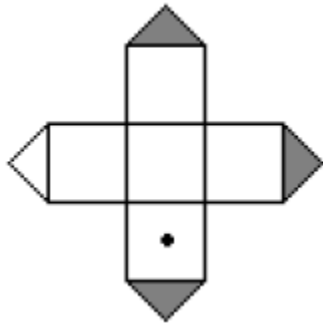


(4)

Ans B

10. The figure given on the left-hand side in each of the following questions is folded to form a box. Choose from the alternatives (1), (2), (3) and (4) the boxes that is similar to the box formed.

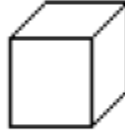
A. 1 & 2 only B. 2 & 4 only C. 2 & 3 only D. 1 & 4 only



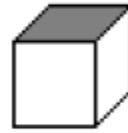
(1)



(2)



(3)



(4)

Ans B

CUBES & CUBOIDS

By:-

**Naveen Kumar Jain
(Aptitude Trainer)**

INTRODUCTION

- ❑ In a cube or a cuboid there are 6 Faces 8 Vertices or corners 12 Edges
 - ❑ In a cube length, breadth and height are same while in cuboid these are different.
 - ❑ In a cube the number of unit cubes = $(\text{side})^3$.
 - ❑ In cuboid the number of unit cube = $(l \times b \times h)$.
 - ❑ To get maximum number of smaller cubes with minimum number of required cuts, we should cut in equal numbers along all three axes.
 - ❑ Let n be the number of smaller cubes along each edge.
 - ❑ Then maximum cubes can be formed = n^3
 - ❑ Minimum number of cuts are required = $(n-1)+(n-1)+(n-1)$ or $3(n-1)$
- 

IMPORTANT FORMULAS FOR CUBE

Number of smaller cubes with 3 faces painted = **8 (Always)**

Number of smaller cubes with 2 faces painted = **$12 \times (n - 2)$**

Number of smaller cubes with 1 face painted = **$6 \times (n - 2)^2$**

Number of smaller cubes with none of the
face painted = **$(n - 2)^3$**

Number of visible cubes from outside = **$n^3 - (n-2)^3$**

PRACTICE QUESTIONS

A cube of each side 4 cm, has been painted black, red and green on pairs of opposite faces. It is then cut into small cubes of each side 1 cm.

1. How many small cubes will be there ?

$$\begin{aligned}\text{Total no. of smaller cubes} &= (\text{sides})^3 \\ &= (4)^3 = 64\end{aligned}$$

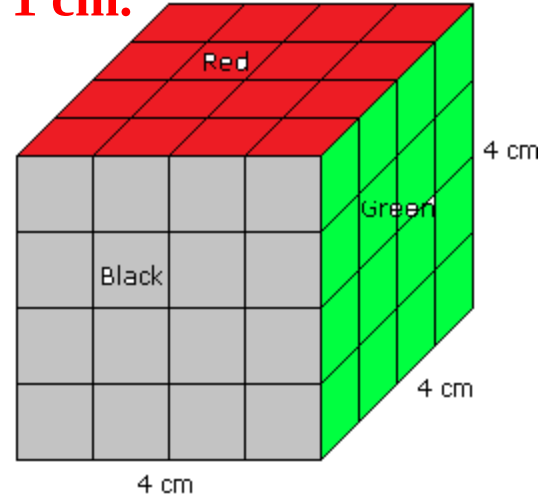
2. How many small cubes will have three faces painted ?

The small cube having three faces colored are situated at the corners of the big cube

As there are 8 corners,
therefore the required number of such cubes is always 8

3. How many small cubes will have only two faces painted ?

$$\begin{aligned}\text{No. of small cubes with two faces colored} &= (n - 2) \times 12 \\ &= (4 - 2) \times 12 \\ &= 24\end{aligned}$$



PRACTICE QUESTIONS

4. How many small cubes will have only one face painted ?

$$\begin{aligned} \text{Number of smaller cubes with 1 face painted} &= 6 \times (n - 2)^2 \\ &= 6 \times (4 - 2)^2 \\ &= 6 \times (4) \\ &= 24 \end{aligned}$$

5. How many small cubes will have no faces painted ?

$$\begin{aligned} \text{Number of smaller cubes with none of the face painted} &= (n - 2)^3 \\ &= (4 - 2)^3 \\ &= 8 \end{aligned}$$

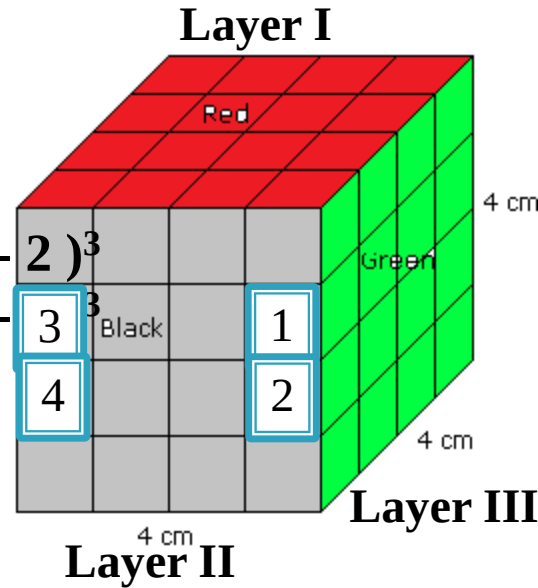
6. How many small cubes will have only two faces painted in black and green and all other faces unpainted ?

There are 4 small cubes in layer II and 4 small cubes in layer III which have two faces painted green and black.

Required no. of such small cubes = 4 + 4 = 8

7. How many small cubes will have only two faces painted in red and =8

**8. How many small cubes will have only two faces painted in red and =8
black and all other faces unpainted ?**



PRACTICE QUESTIONS

9. How many small cubes will have only black painted ?

There will be 8 small cubes which have only black paint.

Four cubes will be form one side and 4 from the opposite side.

10. How many small cubes will have only red painted ? =8

11. How many small cubes will have only green painted ? =8

12. How many small cubes will have at least one face painted ?

No. of small cubes having at least one face painted

= No. of small cubes having 1 face painted
+ 2 faces painted + 3 faces painted

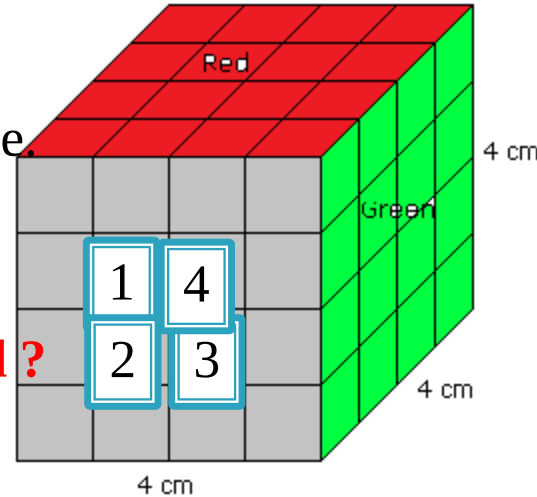
= 24 + 24 + 8

= 56

13. How many small cubes will have at least two face painted ?

No. of small cubes having at least two face painted

No of small cubes 2 faces painted + 3 faces painted
= 24 + 8 = 32



PRACTICE QUESTIONS

The following questions are based on the information given below:

There is a cuboid whose dimensions are $4 \times 3 \times 3$ cm.

The opposite faces of dimensions 4×3 are colored yellow.

The opposite faces of other dimensions 4×3 are colored red.

The opposite faces of dimensions 3×3 are colored green.

Now the cuboid is cut into small cubes of side 1 cm.

1. How many small cubes will have only two faces colored ?

A. 12 B. 24 C. 16 D. 12

2. How many small cubes have three faces colored ?

A. 24 B. 20 C. 16 D. 8

3. How many small cubes will have no face colored ?

A. 1 B. 2 C. 4 D. 8

4. How many small cubes will have only one face colored ?

A. 10 B. 12 C. 14 D. 18

PRACTICE QUESTIONS

The following questions are based on the information given below:

There is a cuboid whose dimensions are $4 \times 3 \times 3$ cm.

The opposite faces of dimensions 4×3 are colored yellow.

The opposite faces of other dimensions 4×3 are colored red.

The opposite faces of dimensions 3×3 are colored green.

Now the cuboid is cut into small cubes of side 1 cm.

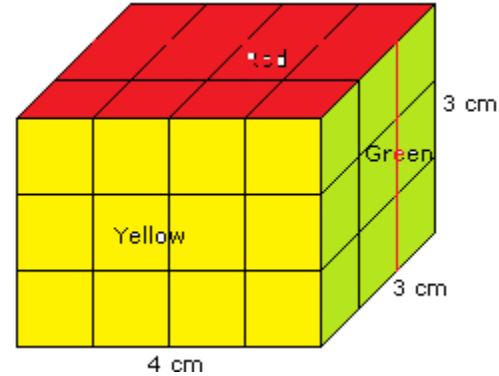
1. How many small cubes will have only two faces colored ?

A. 12

B. 24

C. 16

D. 12



Number of small cubes having only two faces colored

+ 6 from the front
+ 6 from the back
+ 2 from the left
+ 2 from the right

= 16 **Ans C**

PRACTICE QUESTIONS

The following questions are based on the information given below

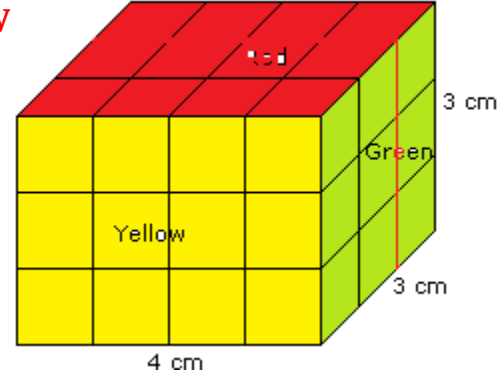
There is a cuboid whose dimensions are $4 \times 3 \times 3$ cm.

The opposite faces of dimensions 4×3 are colored yellow.

The opposite faces of other dimensions 4×3 are colored red.

The opposite faces of dimensions 3×3 are colored green.

Now the cuboid is cut into small cubes of side 1 cm.



2. How many small cubes have three faces colored ?

A. 24

B. 20

C. 16

D. 8

Such cubes are related to the corners of the cuboid and there are 8 corners.

Hence, the required number is 8. **Ans D**

PRACTICE QUESTIONS

The following questions are based on the information given below

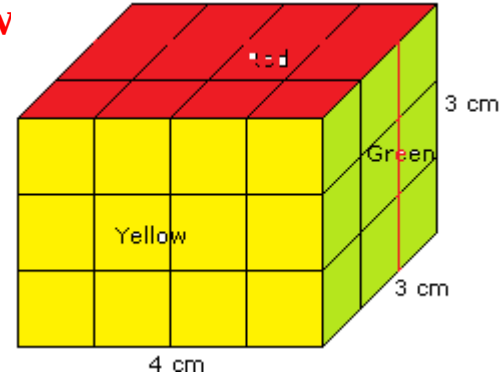
There is a cuboid whose dimensions are 4 x 3 x 3 cm.

The opposite faces of dimensions 4 x 3 are colored yellow.

The opposite faces of other dimensions 4 x 3 are colored red.

The opposite faces of dimensions 3 x 3 are colored green.

Now the cuboid is cut into small cubes of side 1 cm.



3. How many small cubes will have no face colored ?

A. 1

B. 2

C. 4

D. 8

Number of small cubes have no face colored

$$= (4 - 2) \times (3 - 2)$$

$$= (2) \times (1)$$

$$= 2 \quad \text{Ans B}$$

PRACTICE QUESTIONS

The following questions are based on the information given below

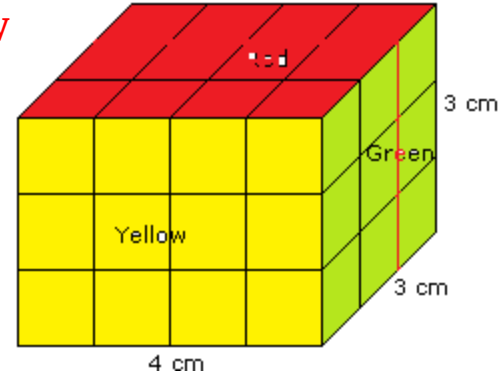
There is a cuboid whose dimensions are $4 \times 3 \times 3$ cm.

The opposite faces of dimensions 4×3 are colored yellow.

The opposite faces of other dimensions 4×3 are colored red.

The opposite faces of dimensions 3×3 are colored green.

Now the cuboid is cut into small cubes of side 1 cm.



4. How many small cubes will have only one face colored ?

A. 10

B. 12

C. 14

D. 18

Number of small cubes having only one face colored $= 2 \times 2 + 2 \times 2 + 2 \times 1$
 $= 4 + 4 + 2$

$= 10$ **Ans A**

1. If you make 4 cuts on one side of a cake, how many pieces do you get ?



2. How many pieces will you get if you make 4 cuts along the length and 4 cuts along the breadth of a solid cube ?



3. If you make 4 cuts along all the three dimensions of a cube, then how many small cubes will you get ?



4. How many cuts should be made to get 64 small cubes out of a cube ?



- 5. A cube of side 4 cm is cut into smaller cubes of side 1 cm.**
- A. How many smaller cubes are painted on two faces ?**
 - B. How many cubes are painted on three faces ?**
 - C. What is the maximum number of cube that can be taken out of the bigger cube and make it a hollow one ?**
- 

Q.6-9 A solid cube of side 8 cm has been painted red, blue and black on pairs of opposite faces. It is then cut into cubical blocks of side 2 cm each.

6. How many cubes have no face painted?

A. 1 B. 4 C. 8 D. 27

7. How many cubes have only one face painted?

A. 8 B. 16 C. 24 D. 28

8. How many cubes have only two face painted?

A. 8 B. 16 C. 20 D. 24

9. How many cubes have three faces painted?

A. 0 B. 4 C. 6 D. 8



Q.10-13 A solid cube of side 8 cm has been painted red, blue and black on pairs of opposite faces. It is then cut into cubical blocks of side 2 cm each.

10. How many cubes have three faces painted with three different colors?

- A. 0 B. 4 C. 8 D. 12**

11. How many cubes have two faces painted red and black and all the other faces unpainted ?

- A. 4 B. 8 C. 16 D. 32**

12. How many cubes have two of their faces painted black?

- A. 2 B. 4 C. 8 D. None**

13. How many cubes have one face painted red and all the other faces unpainted?

- A. 4 B. 8 C. 12 D. 16**
- 

Directions (14-17): There are 2 cubes having different dimensions. 2 opposite faces of a cube are red and rest of it is white. 2 opposite faces of the other cube are yellow and rest of it is red. These 2 cubes are divided separately into 64 and 27 small cubes respectively and then mixed together.

14. How many cubes have only one face painted, in red color?
A. 10 B. 14 C. 18 D. 12
15. How many cubes will have only 2 faces colored, in 2 different colors?
A. 24 B. 16 C. 12 D. 28
16. How many cubes will have 3 faces colored, in three different colors?
A. 9 B. 6 C. 4 D. 0
17. How many cubes will have exactly 2 faces painted and both of them painted in the same color?
A. 36 B. 24 C. 12 D. 18
- 

18. A cube of side 4 cm is divided into 125 small cubes. What will be the length of side of small cube?

A. 0.25 cm

B. 0.8 cm

C. 0.5 cm

D. 1 cm



Directions (19-22): A solid cube is painted red on 2 adjacent sides and black on the sides opposite to the red sides and green on the remaining sides. It is then cut into 64 smaller cubes of equal size.

19. How many cubes have only one side colored?
A. 16 B. 24 C. 28 D. 32
20. How many cubes have at least one side as red?
A. 16 B. 24 C. 28 D. 32
21. How many cubes are there with one side green and the adjacent side either red or black and painted on two sides only?
A. 8 B. 16 C. 24 D. 32
22. How many cubes are there which are red on one side and black on the opposite side?
A. 0 B. 4 C. 8 D. 16
- 